

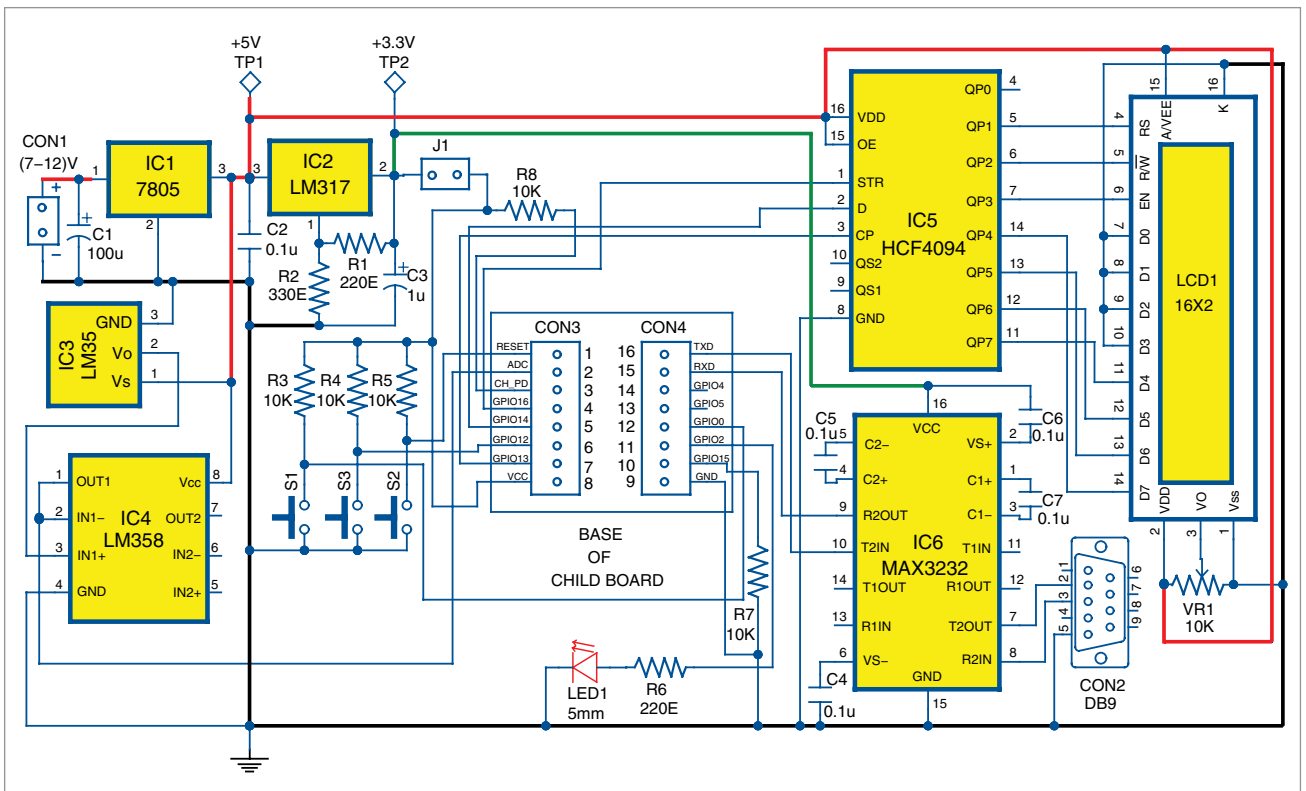


Fig. 3: Circuit diagram of the main board of the low-cost ESP8266 based Wi-Fi Web server

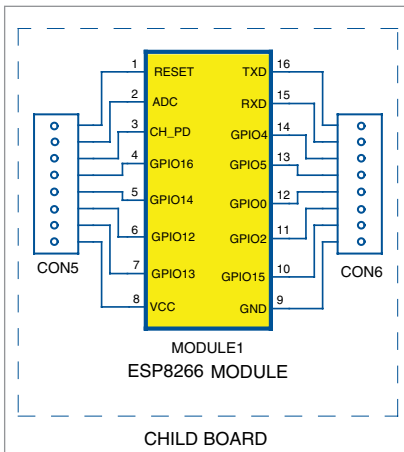


Fig. 4: Circuit diagram of the child board for the circuit shown in Fig. 3

mounted on the main board.

CON2 is used for COM port interface for connecting the module to a PC and the module is programmed using a USB-to-serial converter. If physical COM port is available on the PC, USB-to-serial converter is not necessary.

ESP8266 module has 16 pins. Pin 1 (RESET) is connected to 3.3V

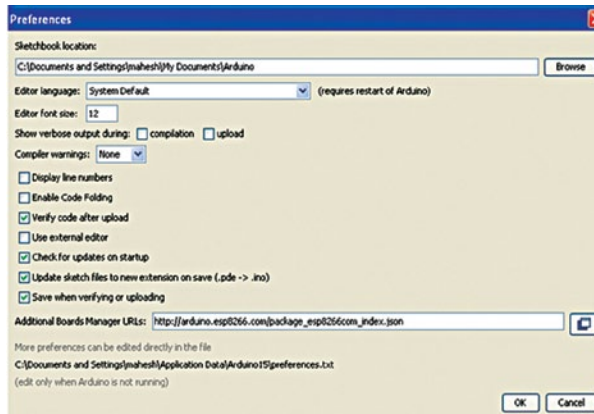


Fig. 5: Preferences window of Arduino IDE

through resistor R5 and push-button S2 is provided for manual reset. Programming mode pin 12 (GPIO0) is connected to 3.3V through R3, and S1 is used to bring the module to programming mode. Pin 3 (CH_PD) is connected to 3.3V through R8.

ESP8266 also integrates a general-purpose 10-bit resolution ADC (pin 2). It is typically used to measure voltage from the sensor or battery. It cannot be used when the chip is

transmitting, otherwise, voltage may be inaccurate.

All digital input/output (I/O) pins are protected from over-voltage with a snap-back circuit connected between the pad and the ground. Snap-back voltage is typically 6V and holding voltage is 5.8V. This provides protection from over-

voltage and ESD. Output devices are also protected from reverse voltage with diodes. LED1 is connected to pin 11 (GPIO2). Pin 6 (GPIO12) is connected to 3.3V through R4 and provided with a push-button (S3) for debugging purposes.

Every supplier of Wi-Fi modules follows a different pattern for output pin arrangement. So no standard PCB pattern can be followed. Readers of EFY can design their own